Proposal for Finance Letter

Extreme Events and Realized Volatility Forecasting: Information Augmentation in Linear and Nonlinear Models

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Abstract—This paper investigates whether intraday extreme-event information improves realized volatility forecasting and whether nonlinear models are better suited to exploit such information. Using high-frequency cryptocurrency data, we construct event-based features capturing the frequency, magnitude, and clustering of extreme losses, including measures derived from Hawkes processes. These features are incorporated into both linear (HAR) and nonlinear (LSTM) forecasting models under a controlled experimental design with identical information sets.
The results show that event-based features provide consistent, though modest, improvements in predictive performance for the HAR model. In contrast, LSTM models do not exhibit systematic gains and, in several cases, performance deteriorates when additional features are included. Further analysis reveals that while exceedance-based measures are strongly correlated with volatility, Hawkes-based intensity captures a distinct, weakly correlated dimension of the data, suggesting complementary but not easily exploitable information.
Overall, the findings indicate that richer information does not necessarily translate into improved forecasts when using more flexible models. Instead, linear models appear more robust in extracting predictive signals from event-based features in this setting.

Keywords—*Realized volatility; HAR model; LSTM; Extreme events; Hawkes processes; Cryptocurrency markets; Volatility forecasting*

1. Introduction

Forecasting realized volatility (RV) is a central problem in financial econometrics due to its direct implications for risk management, derivative pricing, and portfolio allocation. The availability of high-frequency data has enabled the construction of realized measures that provide accurate ex-post estimates of latent volatility, thereby significantly improving empirical modeling and forecasting frameworks [5], [6], [7].

Among the most successful approaches, the Heterogeneous Autoregressive (HAR) model [8] has emerged as a simple yet powerful benchmark [20]. By combining daily, weekly, and monthly components, HAR effectively captures the long-memory behavior of volatility implied by the Heterogeneous Market Hypothesis [4]. Despite its strong empirical performance and interpretability, the model relies exclusively on past realized volatility and does not explicitly incorporate information contained in intraday market dynamics.

However, financial markets—particularly cryptocurrency markets—are characterized by frequent extreme price movements, clustering of events, and strong endogenous dynamics [21], [22], [23]. Empirical evidence suggests that such extreme events occur regularly and exhibit temporal dependence, indicating that volatility may be partially driven by event-based mechanisms rather than solely by aggregated past volatility measures. This raises a natural question: whether information extracted from intraday extreme events can improve volatility forecasting.

A growing strand of the literature has explored the role of jumps and extreme movements in volatility prediction [14], [24], [25]. Existing approaches show that incorporating jump indicators or jump intensity measures can enhance forecasting performance. Nevertheless, these contributions are typically implemented within linear frameworks, which may not fully capture the complex and nonlinear relationships between event dynamics and future volatility.

At the same time, advances in machine learning have introduced flexible models capable of capturing nonlinear dependencies and high-dimensional interactions. In particular, recurrent neural networks such as Long Short-Term Memory (LSTM) architectures [3] have been successfully applied to volatility forecasting problems [16], [17], [25], [26]. Yet, the empirical gains from these models remain difficult to interpret: improvements may arise from model flexibility rather than from the effective use of new information.

This paper adopts an information-centric perspective to address this issue. Instead of focusing solely on model comparison, we investigate whether event-driven features extracted from intraday data provide incremental predictive value for realized volatility, and whether this value depends on the modeling framework used to incorporate them.

Specifically, we construct measures of extreme-event activity using intraday returns and augment them with model-based representations derived from self-exciting point processes. These features are then incorporated into both linear and nonlinear forecasting models, allowing for a controlled comparison of their effectiveness.

The main research question is therefore:

Can intraday extreme-event information improve realized volatility forecasting, and do nonlinear models exploit this information more effectively than linear models?

To answer this question, we design a structured empirical framework that progressively enriches the information set available to the models. This allows us to disentangle three key effects: (i) the value of extreme-event detection, (ii) the contribution of model-based event dynamics, and (iii) the interaction between information richness and model nonlinearity.

Contribution. The contribution of this paper is threefold. First, we provide a systematic evaluation of event-based features in volatility forecasting, distinguishing between frequency, magnitude, and clustering effects. Second, we compare linear and nonlinear models under identical information sets, allowing us to isolate whether predictive improvements arise from richer information or from model flexibility. Third, we provide empirical evidence on the role of event-driven dynamics in cryptocurrency markets, highlighting that additional information does not necessarily translate into improved performance for nonlinear models.

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Relation to the literature. This study relates to three strands of the literature. First, it contributes to realized volatility modeling by extending standard HAR-type frameworks with intraday-derived features. Second, it connects to the literature on jump-based and event-driven volatility forecasting, where extreme events and their intensities have been shown to contain predictive information. Third, it contributes to the growing body of work comparing machine learning and econometric approaches in finance, by providing a controlled setting in which both model classes are evaluated under identical information structures.

Paper structure. The remainder of the paper is organized as follows. Section 2 describes the data and feature construction. Section 3 presents the modeling framework. Section 4 reports the empirical results. Section 5 concludes.

2. Data and features

2.1. Asset Selection and RV construction

Bitcoin (BTC) and Ethereum (ETH) are selected as the primary assets of analysis due to their dominant roles in the cryptocurrency market, both in terms of market capitalization and trading activity. As the two most liquid and widely traded cryptocurrencies, they provide a representative setting for studying volatility dynamics in crypto markets.

Intraday price data are obtained from the Binance API for the BTC/USDT and ETH/USDT trading pairs, sampled at a 5-minute frequency over the period from August 17, 2017, to December 31, 2025, covering a total of 3,059 trading days.[1] This frequency yields 288 observations per day and represents a standard compromise between capturing high-frequency information and mitigating microstructure noise.

Let $P_{t,i}$ denote the price at day t and intraday interval i, with $i = 1, \dots, 288$. Intraday log-returns are defined as

$$r_{t,i} = \log\left(\frac{P_{t,i}}{P_{t,i-1}}\right),$$

and daily realized volatility is computed as

$$\mathrm{RV}_t = \sum_{i=1}^{288} r_{t,i}^2. \tag{1}$$

The dataset is largely complete, with no missing trading days. A small number of days (35) exhibit missing intraday observations. For these cases, realized volatility is adjusted by scaling the sum of squared returns proportionally to the full-day frequency, ensuring comparability across days.

Following standard practice in the literature (e.g., [7], [19]), a square-root transformation is applied to realized volatility to reduce skewness and improve its suitability for linear modeling. Specifically, we define

$$y_t = \sqrt{\mathrm{RV}_t},$$

and, for notational simplicity, refer to y_t as realized volatility throughout the remainder of the paper.

2.2. Intraday information

This section describes the procedure used to identify extreme intraday events and construct the associated features.

2.2.1. Extreme event timestamps

A central challenge in modeling high-frequency financial data is the definition of extreme events. Extreme Value Theory (EVT) provides a natural framework for this purpose, with the Peaks-over-Threshold (POT) approach being one of the most widely used methods. While alternative approaches exist, POT-based methods are preferred due to their simplicity and strong empirical performance in financial applications (e.g., [10], [11], [12]).

In this study, we follow the methodology proposed in [18], where extreme events are identified through a dynamic threshold estimated via quantile regression. This approach allows the threshold to adapt to local market conditions and intraday patterns.

To focus on downside risk, we define

$$X_{t,i} = -r_{t,i},$$

so that large positive values of $X_{t,i}$ correspond to extreme losses. Conditional on the available information set $\mathcal{H}_t^*$, the time-varying threshold is specified as

$$\begin{aligned} u_{t,i} := u_{t,i}(\tau \mid \mathcal{H}_t^*) &= \omega(\tau) + \alpha_1(\tau)|X_{t,i-1}| + \alpha_2(\tau)|X_{t,i-2}| \\ &+ \sum_{j=1}^{3}\left[\varphi_j(\tau)\cos\left(2\pi j\frac{i}{D}\right) + \psi_j(\tau)\sin\left(2\pi j\frac{i}{D}\right)\right], \end{aligned} \tag{2}$$

where $i = 1, \dots, D$ indexes the intraday intervals and $D = 288$ corresponds to the number of 5-minute observations per day.

The threshold is estimated as the solution to a quantile autoregressive (QAR) model, obtained by minimizing the pinball loss function associated with quantile level τ (see [18] for details). In this study, we set $\tau = 0.95$, following standard recommendations in the EVT literature (e.g., [10]). This choice focuses on the upper tail of the loss distribution while retaining a sufficient number of observations for reliable estimation.

The information set $\mathcal{H}_t^*$ consists of intraday observations from the previous seven days, excluding day t. This rolling-window specification ensures that the threshold is constructed in a strictly causal manner, avoiding look-ahead bias and aligning the procedure with the forecasting framework of the study. The choice of a seven-day window reflects a trade-off between statistical reliability and adaptability: given the 5-minute sampling frequency ($D = 288$ observations per day), this window provides a sufficiently large sample to estimate the quantile regression model while remaining responsive to local market dynamics.

[1] The Python code used for data extraction and preprocessing is available upon request.

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The specification incorporates two key components. First, lagged absolute returns capture short-term dependence in extreme movements. Second, the Fourier expansion accounts for intraday periodicity, a well-documented feature of high-frequency financial data. Together, these elements allow the threshold to flexibly adjust across both time and market conditions.

An observation is classified as an extreme event whenever $X_{t,i} > u_{t,i}$, yielding a sequence of event timestamps that form the basis for subsequent feature construction.

2.2.2. Event-based features

To incorporate intraday extreme-event information into daily forecasting models, it is necessary to aggregate the detected events into daily-level features. We construct two complementary measures capturing the frequency and magnitude of extreme losses.

Frequency of events. Let $\mathbf{1}_{\{X_{t,i}>u_{t,i}\}}$ denote the indicator function for an extreme event. The total number of downside events on day t is defined as

$$N_t = \sum_{i=1}^{D} \mathbf{1}_{\{X_{t,i}>u_{t,i}\}},$$

where $D = 288$ is the number of intraday intervals. To obtain a scale-free measure, we define the daily event ratio as

$$R_t = \frac{N_t}{D}.$$

This metric captures the frequency of extreme losses within a given day and can be interpreted as a simple proxy for event intensity. However, it does not account for the magnitude of such events.

Exceedance-based metric. To incorporate information on the severity of extreme movements, we define the exceedance process

$$m_{t,i} = \max\left(0, X_{t,i} - u_{t,i}\right),$$

which measures the extent to which observations exceed the threshold.

We then construct the daily exceedance metric as

$$E_t = \sqrt{\sum_{i=1}^{D} m_{t,i}^2}.$$

This measure is analogous in spirit to realized volatility, but restricted exclusively to extreme losses. As such, it captures the magnitude of tail events while ignoring non-extreme observations.

While E_t reflects the intensity of extreme losses in terms of size, it does not capture their temporal distribution or clustering within the day.

2.2.3. Hawkes intensity

While the previously defined measures capture the frequency and magnitude of extreme events, they do not account for their temporal dependence. Financial markets are known to exhibit clustering of extreme movements, a feature naturally modeled by self-exciting point processes.

Hawkes processes, introduced by [1] and widely applied in finance (see, e.g., [13]), provide a flexible framework to capture such endogenous dynamics. Let $N(t)$ denote the counting process of extreme events up to time t. The conditional intensity function is defined as

$$\lambda(s \mid \mathcal{H}_s) = \lim_{\Delta \to 0} \mathbb{E}\left[\frac{N(s+\Delta) - N(s)}{\Delta} \,\middle|\, \mathcal{H}_s\right], \tag{3}$$

which represents the instantaneous expected arrival rate of events given the past history $\mathcal{H}_s$.

Under the exponential kernel specification, the intensity takes the form

$$\lambda(s) = \mu + \sum_{t_i < s} \alpha \exp\left(-\beta(s - t_i)\right), \tag{4}$$

where μ is the baseline intensity, α controls the magnitude of self-excitation, and β governs the decay rate of past events. This specification captures the clustering behavior of extreme events through the excitation term.

Parameter estimation is performed via maximum likelihood, following the framework of [2]. In practice, to ensure numerical stability and tractability, the decay parameter β is fixed, and the remaining parameters are estimated using the `tick` library.[2]

To obtain a daily feature, the Hawkes process is fitted using a rolling window of the previous 30 days, excluding day t, ensuring a strictly causal estimation. The fitted model is then used to compute the intensity over the intraday grid of day t.

Formally, the integrated intensity over day t is defined as

$$\Lambda_t = \int_{\text{day } t} \lambda(s)\, ds, \tag{5}$$

which represents the expected number of events over the day under the fitted model. In practice, this quantity is approximated on the discrete intraday grid. We define the empirical feature as the average intensity,

$$\bar{\Lambda}_t = \frac{1}{D} \sum_{i=1}^{D} \lambda(s_{t,i}), \tag{6}$$

where $s_{t,i}$ denotes the timestamp of the i-th intraday interval.

[2]The `tick` library provides efficient implementations for Hawkes process estimation; see **tick** for details.

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This measure captures the endogenous clustering of extreme events, reflecting how past occurrences increase the likelihood of future events. Unlike the event count and exceedance metrics, $\bar{\Lambda}_t$ incorporates information on the temporal structure of events, but does not account for their magnitude. Extensions based on marked point processes could incorporate this additional dimension.

2.2.4. Information decomposition

To evaluate the incremental value of event-based information, we adopt a structured feature construction strategy based on progressively enriching the information set available to the models. This approach allows us to disentangle the individual and joint contributions of frequency, magnitude, and clustering effects in volatility forecasting.

Starting from the baseline specification using only past realized volatility y_t, we sequentially incorporate the proposed features. Specifically, we consider the following hierarchy of information sets:

$$[y_t] \to [y_t, R_t] \to [y_t, E_t] \to [y_t, \bar{\Lambda}_t] \to [y_t, R_t, E_t] \to [y_t, \bar{\Lambda}_t, E_t] \to [y_t, R_t, E_t, \bar{\Lambda}_t].$$

This design serves three main purposes. First, it allows us to assess the marginal contribution of each feature independently. Second, it enables the evaluation of interaction effects between frequency, magnitude, and clustering information. Third, it provides a consistent framework to compare linear and nonlinear models under identical information sets.

By structuring the empirical analysis in this way, we can determine whether improvements in forecasting performance arise from the inclusion of new information, from the model's ability to exploit such information, or from the interaction between both.

3. Models

3.1. Econometric baseline: HAR

The Heterogeneous Autoregressive (HAR) model, introduced by [8] and motivated by the Heterogeneous Market Hypothesis [4], serves as the benchmark specification. The model captures long-memory dynamics in volatility through the aggregation of daily, weekly, and monthly components.

Using the square-root transformation of realized volatility, $y_t = \sqrt{\mathrm{RV}_t}$, the HAR model is specified as

$$y_{t+1} = \beta + \beta_d y_t^d + \beta_w y_t^w + \beta_m y_t^m, \tag{7}$$

where

$$y_t^w = \frac{1}{7}\sum_{i=1}^{7} y_{t-i}, \qquad y_t^m = \frac{1}{30}\sum_{i=1}^{30} y_{t-i},$$

reflecting the 24/7 trading structure of cryptocurrency markets.

To incorporate additional information, the model is extended to include exogenous covariates. For a generic daily feature C_t, we define analogous aggregated components $C_t^d = C_t$, $C_t^w = \frac{1}{7}\sum_{i=1}^{7} C_{t-i}$, and $C_t^m = \frac{1}{30}\sum_{i=1}^{30} C_{t-i}$, and augment the HAR specification accordingly.

The model is estimated via ordinary least squares (OLS). While alternative estimation methods such as weighted least squares (WLS) have been shown to improve performance ([19]), OLS is adopted as the baseline for comparability.

The main advantage of the HAR model lies in its interpretability and ability to capture long-memory effects with a simple linear structure. However, its linear specification may limit its ability to fully exploit complex, nonlinear relationships in the data.

3.2. LSTM architecture

To capture nonlinear dynamics, we consider Long Short-Term Memory (LSTM) networks [3]. LSTM models process sequential inputs and learn temporal dependencies through a hidden state representation.

Let $h_t \in \mathbb{R}^H$ denote the hidden state at time t. Given an input sequence, the LSTM produces a representation z_t, which is used to generate predictions

$$\hat{y}_{t+1} = f(z_t),$$

where $f(\cdot)$ is a feedforward mapping. In the baseline specification, f is implemented as a linear layer, as more complex alternatives such as multilayer perceptrons did not yield consistent improvements in out-of-sample performance.

The input to the model consists of sequences of length $L = 30$, constructed as

$$\mathbf{X}_t \in \mathbb{R}^{30\times(1+N_c)},$$

where each row contains $[y_{t-i}, C_{t-i}]$ and N_c denotes the number of covariates. The choice of $L = 30$ aligns the information set with the HAR model, allowing for a fair comparison between linear and nonlinear approaches. While HAR imposes a fixed linear aggregation structure, the LSTM is free to learn flexible nonlinear representations from the same input information.

Model hyperparameters include the hidden dimension H, number of layers, and the choice of representation z_t (e.g., last hidden state). Empirically, shallow architectures were found to be sufficient, as increasing depth did not lead to systematic performance gains. Similarly, small hidden dimensions (e.g., $H = 8, 16$) resulted in underfitting, while moderate sizes provided more stable performance.

Training is performed using the Adam optimizer, as alternatives such as AdamW did not produce meaningful differences in predictive accuracy. Early stopping based on validation loss is employed to prevent overfitting and was found to be essential for stable training.

Finally, different output specifications were evaluated. While nonlinear activation functions such as ReLU or softplus can be used to enforce positivity, they did not improve performance in our setting, and a linear output layer was retained in the baseline specification.

4. Experiments and results

Let $(y_t)_t$ denote the observed series of realized volatility and $(\hat{y}_t)_t$ the corresponding predictions. Model evaluation is conducted using a rolling window scheme.

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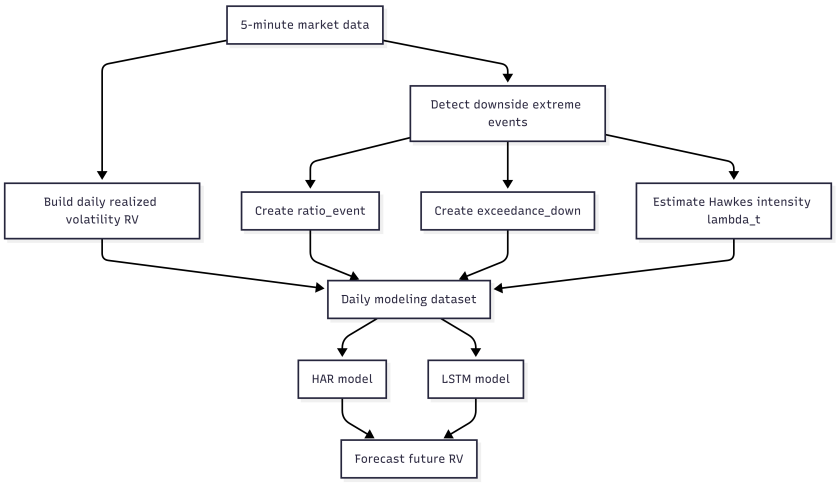

Figure 1. Data processing pipeline, including intraday return construction, extreme-event detection, feature aggregation, and model input preparation.

Specifically, each iteration consists of 1,200 days of training data and 300 days of test data. For LSTM models, the training set is further split into 900 days for training and 300 days for validation. Early stopping is employed based on validation loss to prevent overfitting, with a fixed patience parameter. A sufficiently large number of training epochs is allowed to ensure convergence.

This rolling-window framework ensures a realistic out-of-sample evaluation and allows for consistent comparison across model specifications and information sets.

Evaluation metrics. Forecast accuracy is assessed using Mean Squared Error (MSE) and the QLIKE loss function [9]. The MSE is defined as

$$\mathrm{MSE} = \frac{1}{T}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2 . \tag{8}$$

Since the target variable is defined as $y_t = \sqrt{\mathrm{RV}_t}$, the QLIKE loss is computed on the variance scale. Specifically, letting $\widehat{\mathrm{RV}}_t = \hat{y}_t^2$, the QLIKE loss is given by

$$\mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{\mathrm{RV}_t}{\widehat{\mathrm{RV}}_t} - \log\left(\frac{\mathrm{RV}_t}{\widehat{\mathrm{RV}}_t}\right) - 1\right). \tag{9}$$

QLIKE is particularly suitable for volatility forecasting, as it provides a robust loss function under noisy volatility proxies.

Data pipeline. The full data processing and feature construction pipeline is summarized in Figure 1, illustrating the transformation from intraday returns to daily features and model inputs.

Model behavior. During hyperparameter tuning of the LSTM models, it was observed that many configurations tend to produce nearly constant predictions over the evaluation window. This behavior suggests a tendency toward mean reversion or variance collapse, where the model approximates the unconditional mean of the target variable rather than capturing its dynamics.

Several potential explanations were considered. First, underfitting was ruled out, as increasing the number of epochs and applying early stopping did not improve performance. Second, the predictive signal in past volatility is known to be strong, as evidenced by the HAR benchmark. Therefore, the observed behavior likely reflects the difficulty of LSTM models in extracting predictive structure from the available features under the given sample size and noise conditions.

Main results. Table **??** reports the changes in QLIKE loss relative to baseline models without event-based features. The results show that augmenting the HAR model with extreme-event features leads to consistent, albeit modest, improvements in predictive performance across both Bitcoin and Ethereum.

In contrast, the LSTM models do not exhibit systematic gains from the inclusion of additional features. In several cases, the introduction of event-based variables increases the forecast error, and performance deteriorates further when combining multiple features. While isolated improvements are observed for specific configurations, these effects are not robust.

Overall, the results suggest that the predictive value of extreme-event features is limited and depends critically on the modeling framework. In particular, linear models appear to extract more stable benefits from these features than nonlinear architectures.

Feature relationships. To further understand these results, Figures 2 and 3 present correlation matrices between the constructed features and realized volatility.

The exceedance-based metric E_t exhibits a strong positive correlation with y_t (approximately 0.8), reflecting its construction as a tail-restricted analogue of realized volatility. The event ratio R_t shows moderate correlation (around 0.4–0.5), indicating that frequency information provides a partial signal of volatility dynamics.

In contrast, the Hawkes-based feature $\bar{\Lambda}_t$ displays near-zero linear correlation with both y_t and the other features. This suggests that it captures a distinct dimension of the data, related to temporal clustering rather than magnitude or frequency.

Interestingly, despite its low correlation, $\bar{\Lambda}_t$ contributes to performance improvements in the HAR model when combined with other features. This indicates that its informational content may be nonlinear or complementary, rather than directly predictive in a linear sense.

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5. Conclusion

5.1. Summary of findings

This study investigates whether intraday extreme-event information improves realized volatility forecasting and whether nonlinear models are better suited to exploit such information.

The empirical results show that augmenting the HAR model with event-based features leads to consistent, albeit modest, improvements in predictive performance. In contrast, LSTM models do not exhibit systematic gains from the inclusion of additional features. Across most specifications, LSTM predictions tend to collapse toward nearly constant values, suggesting difficulties in extracting meaningful predictive structure from the data.

Importantly, the HAR model remains robust to the inclusion of additional regressors, even when their individual predictive contribution is weak or not linearly correlated with the target variable. Conversely, the performance of LSTM models deteriorates in several cases when incorporating additional features, particularly when combining multiple sources of information.

Overall, these findings suggest that the predictive value of extreme-event features is limited and that, in this setting, linear models provide more stable and reliable performance than their nonlinear counterparts.

5.2. Limitations and extensions

Several limitations of this study should be acknowledged. First, the analysis considers a limited set of model specifications, including a baseline HAR model and standard LSTM architectures. Alternative configurations, such as weighted least squares (WLS) estimation for HAR or different optimization schemes (e.g., AdamW) for LSTM, may yield different results.

Second, the construction of extreme-event features involves multiple design choices, including the jump detection method, threshold specification, and Hawkes process calibration. These elements introduce additional degrees of freedom that may influence the results and warrant further robustness analysis.

Third, the empirical evaluation is restricted to a small set of cryptocurrencies. Extending the analysis to additional assets and financial markets would provide a more comprehensive assessment of the findings.

Finally, formal statistical validation of forecast differences, such as Diebold–Mariano tests or Model Confidence Set procedures, is left for future work.

5.3. Future work

Several directions for future research emerge from this study. First, incorporating interpretable machine learning mechanisms, such as attention-based LSTM architectures, may provide insights into how past observations contribute to volatility forecasts.

Second, continuous-time neural architectures, such as the Neural Hawkes process [15], could be used to directly integrate intraday event dynamics into the forecasting framework.

Third, extending the analysis to include alternative event definitions, such as positive jumps or signed measures, may improve the representation of extreme market movements.

Finally, further investigation is needed to ensure fairness in information usage across models. While HAR relies on aggregated features, LSTM models operate on full sequences, which may introduce differences in how information is processed and exploited.

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5.5. Disclaimer

The author declares no affiliation with any financial institution and no financial interest related to this research. The results and opinions expressed in this paper are solely those of the author. Artificial intelligence tools were used as writing and coding assistants; all research design, analysis, and interpretation were conducted by the author.

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6. Annex

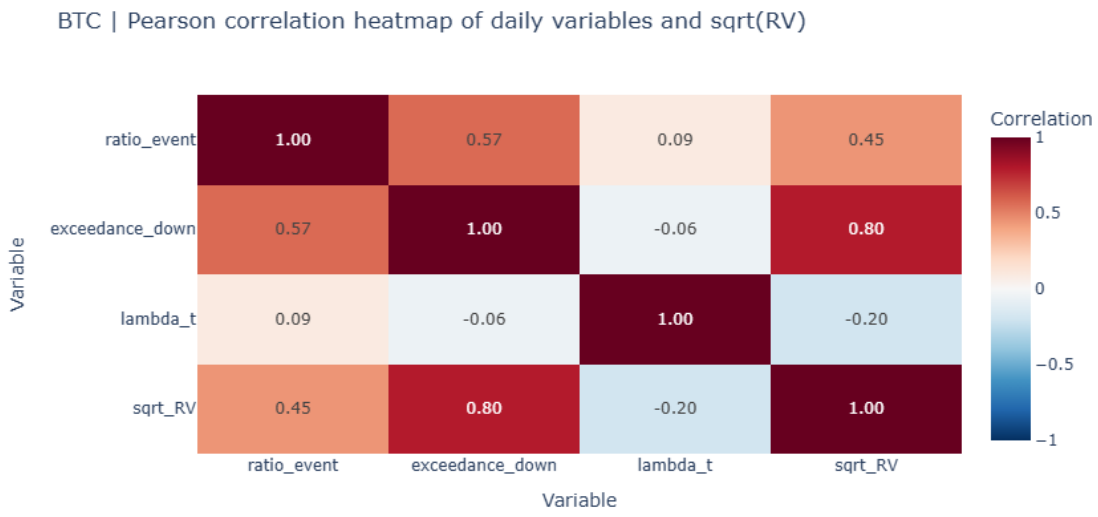

Figure 2. Correlation matrix of features for Bitcoin (BTC).

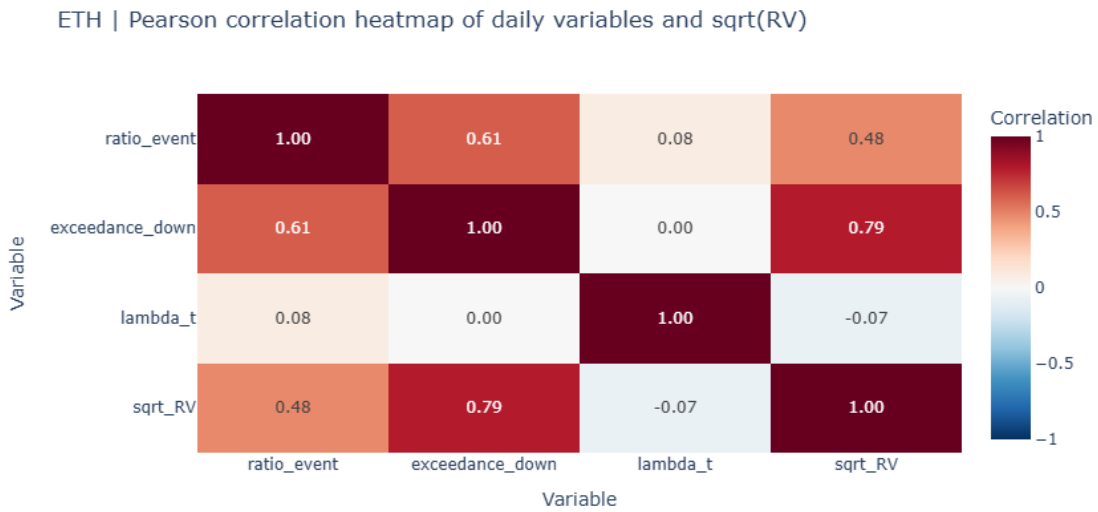

Figure 3. Correlation matrix of features for Ethereum (ETH).

Features Model	None	R_t	E_t	$\bar{\Lambda}_t$	$[R_t, E_t]$	$[E_t, \bar{\Lambda}_t]$	$[R_t, E_t, \bar{\Lambda}_t]$
HAR-baseline	0.0000	-0.0053	-0.0111	-0.0067	-0.0163	-0.0155	-0.0181
LSTM-hs32	0.0000	0.0130	-0.0010	-0.0029	0.0028	0.0034	0.0237
LSTM-hs64	0.0000	0.0126	0.0086	0.0156	0.0201	0.0365	0.0143

Table 1. Changes in QLIKE loss for Bitcoin (BTC) relative to the baseline specification without event-based features. Negative values (green) indicate improvements in predictive performance, while positive values (red) indicate deterioration. Results are reported across different model specifications and progressively enriched information sets.

Features Model	None	R_t	E_t	$\bar{\Lambda}_t$	$[R_t, E_t]$	$[E_t, \bar{\Lambda}_t]$	$[R_t, E_t, \bar{\Lambda}_t]$
HAR-baseline	0.0000	-0.0042	-0.0039	0.0015	-0.0065	0.0004	-0.0017
LSTM-hs64	0.0000	0.0044	0.0243	-0.0079	0.0236	0.0120	0.0376
LSTM-hs32	0.0000	-0.0106	0.1586	0.0015	0.0042	0.0178	0.0067

Table 2. Changes in QLIKE loss for Ethereum (ETH) relative to the baseline specification without event-based features. Negative values (green) indicate improvements in predictive performance, while positive values (red) indicate deterioration. Results are reported across different model specifications and progressively enriched information sets.

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